

Burn/CO/CN Case



Section I: Scenario Demographics

Scenario Title:	30% BSA Burn with CO/CN Poisoning		
Date of Development:	16/05/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	kcaners@gmail.com

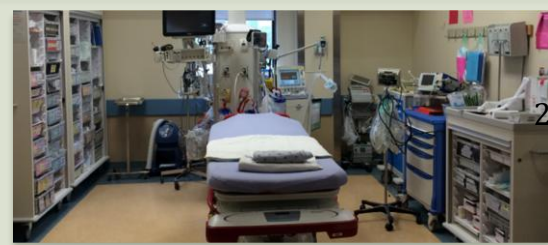
Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to the complex management and differential considerations in a critically ill patient from a house fire.
CRM Objectives:	Effectively lead team members through complex critical scenario.
Medical Objectives:	1) Recognize the need to intubate patient with significant burns. 2) Adequately fluid resuscitate patient with significant burns. 3) Recognize possible CO and CN exposure and administer treatment.

Case Summary: Brief Summary of Case Progression and Major Events
A 33 year-old female is dragged out of a burning house and presents to the ED unresponsive. She has soot on her face, singed eyebrows, and burns to her entire chest, the front of her right arm, and part of her right leg. She is hypotensive and tachycardic with a GCS of 3. The team should proceed to intubate and fluid resuscitate. After this, the team will receive a critical VBG result that reveals profound metabolic acidosis, carboxyhemoglobin of 25 and a lactate of 11. If the potential for cyanide toxicity is recognized and treated, the case will end. If it is not, the patient will proceed to VT arrest.

Facilitators Required to Run Session
Instructors (faculty or senior resident): 1-2 (one to observe, one to play family member). Could easily have observing faculty member come in to play family member at end of case if short on instructors.
Confederate nurse: 1 (to assist at bedside, cue as to patient appearance)
Sim tech: 1

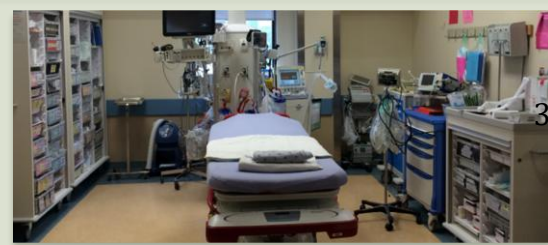
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates		Brief Description of Role	
Family member		Arrives part way through scenario very distraught about fire. When questioned, will eventually give history that the patient has recently been depressed. Has previously tried hanging herself.	
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask on patient at start of case	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Soot on face, singed eyebrows. Fake burns to chest, right arm/abdomen/leg. (Nothing circumferential) Cutting scars across left forearm. Non-rebreather on patient at start of case.			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	12 min
		Debriefing:	10 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 33-year-old female has just been brought into your tertiary care ED. She was dragged out of a house fire and is unresponsive. The etiology of the fire is unclear, but the home was severely damaged. The EMS crew that transported her noted significant burns across her chest, abdomen, arm, and leg.

B. Patient Profile and History

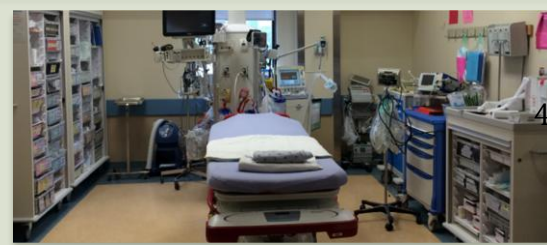
Patient Name: Sheena Sherman	Age: 33	Weight: 70
Gender: ☐ M ☒ F	Code Status: Full	
Chief Complaint: burns, aLOC		
History of Presenting Illness: Dragged out of a house fire unresponsive. No other members of household at the scene. Unclear etiology of fire.		
Past Medical History:	Unknown	Medications: Unknown
Allergies: None known.		
Social History: Unknown		
Family History: Unknown.		

Review of Systems:	CNS:	Unable
	HEENT:	Unable
	CVS:	Unable
	RESP:	Unable
	GI:	Unable
	GU:	Unable
	MSK:	Unable
INT:	Unable	

C. Baseline Simulator State and Physical Exam

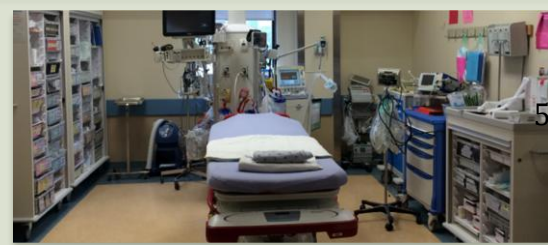
☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 130/min	BP: 75/45	RR: 18/min
O ₂ SAT: 96%		
Rhythm: sinus tach	T: 36.1°C	Glucose: 6.1 mmol/L
GCS: 3 (E 1 V 1 M1)		
General Status: Unresponsive.		
CNS:	GCS 3. Pupils 3mm bilat, minimally reactive.	
HEENT:	No signs HI. Soot on face. Singed eyebrows.	
CVS:	Nil.	
RESP:	GAEB. No adventitious.	
ABDO:	Abdo soft, NT.	
GU:	Nil.	
MSK:	No signs trauma.	Burns to entire chest, R arm, abdomen, and R leg. Cutting scars to L arm.

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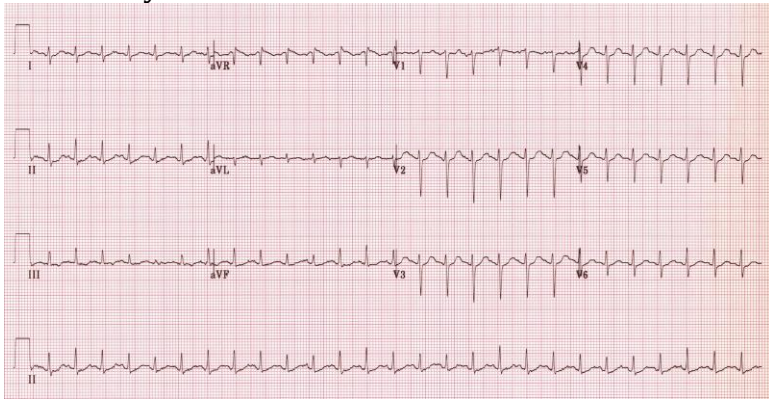
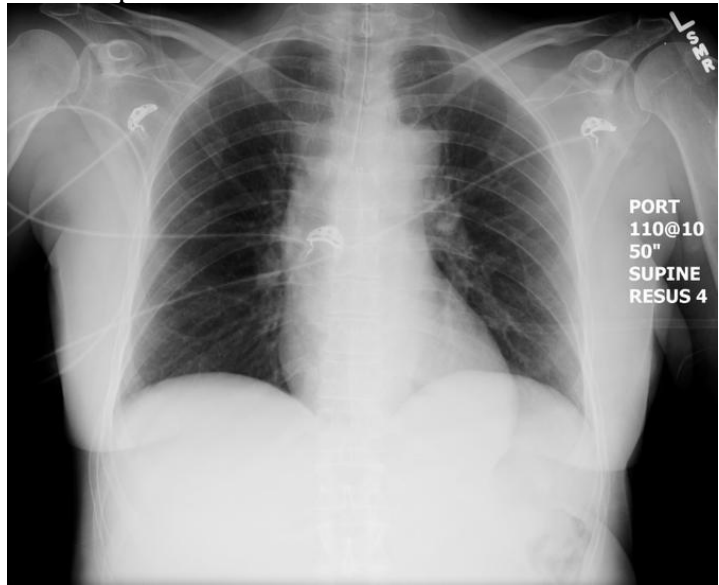
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 130/min BP: 75/45 RR: 18/min O ₂ SAT: 96 % T: 36.1°C	Unresponsive with GCS 3.	<u>Learner Actions</u> - ☐ 2 large bore IVs, bolus 2L - ☐ 100% O ₂ , monitors - ☐ Labs: VBG, carboxyHb, lactate, coags, trop, lytes, Cr - ☐ Check glucose: 6.3 - ☐ Td administration - ☐ Full exposure, determine burn area (22.5% TBSA)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - 2L IVF → HR 120, BP 82/50 - If no discussion of intubation by 5 min, RN to prompt <u>Triggers</u> <i>For progression to next state</i> - Intubate → 2. Intubation - 7 minutes → 3. Critical VBG
2. Intubation HR → 120 BP → 80/50	Unchanged.	<u>Learner Actions</u> - ☐ Push dose pressors at bedside - ☐ Use BP neutral agent (ketamine or etomidate) - ☐ Double set-up (possible difficult airway) - ☐ Intubate - ☐ CXR to confirm placement	<u>Modifiers</u> - Propofol used → BP 60/30 - Any other agent used → BP 70/45 <u>Triggers</u> - 7 minutes → 3. Critical VBG
3. Critical VBG HR → 115 BP → 75/45		<u>Learner Actions</u> - ☐ Ensure patient on 100% O ₂ - ☐ Continue IVF bolus - ☐ Start vasopressor infusion - ☐ Give hydroxycobalamin 5mg iv over 15 min - ☐ ± Call Poison Control (they advise hydroxycobalamin) - ☐ ± Na thiosulfate 12.5g iv	<u>Modifiers</u> - Vasopressor started → BP 85/55 <u>Triggers</u> - 10 min → 4. Arrest - Hydroxycobalamin given → 5. Resolution
4. Arrest Rhythm → PEA BP → -/-	Patient pulseless.	<u>Learner Actions</u> - ☐ Ensure quality CPR - ☐ Intubate if not yet done - ☐ Epinephrine q3min - ☐ HCO ₃ ampules - ☐ Give hydroxycobalamin	<u>Modifiers</u> <u>Triggers</u> - Hydroxycobalamin given → 5. Resolution - 12 minutes → 5. Resolution
5. Resolution Rhythm → sinus tach HR → 110 BP → 95/60	Patient remains unresponsive. Family member arrives.	<u>Learner Actions</u> - ☐ Call ICU/Plastics - ☐ Call Poison Centre - ☐ Call for hyperbarics - ☐ Disclose bad news to family	<u>ICU arrives to manage patient</u> END CASE

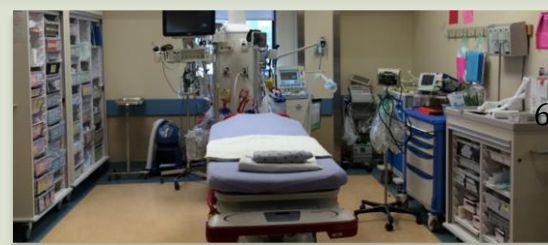


Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results					
VBG	pH: 6.98	PCO ₂ : 28	PO ₂ : 70	HCO ₃ : 6	Lactate: 11
Carboxyhb: 0.4					

Images (ECGs, CXRs, etc.)	
Sinus tachycardia ECG  ECG source: https://lifeinthefastlane.com/ecg-library/sinus-tachycardia/	Normal post-intubation CXR  CXR source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg

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Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video
Objectives			
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Medical Objectives:	1) Recognize the need to intubate patient with significant burns. 2) Adequately fluid resuscitate patient with significant burns. 3) Recognize possible CO and CN exposure and administer treatment.		
Sample Questions for Debriefing			
1) What were the clues to CN toxicity in this patient? 2) What are the recommended treatments for CN toxicity? 3) What is the Parkland formula? How would it be applied in this case? (Hint: it's not) 4) What is the management of CO toxicity? What are the criteria for hyperbarics? 5) What was challenging about this case? 6) How do you feel your team did in working through uncertainty around the diagnosis? Did the leader facilitate input from others?			
Key Moments			
Recognition of need to intubate			
Realization that shock is not responding to fluids			
Recognition of CO and CN poisoning			